

# Autonomous Weed Control Using Drones & Robotics

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**Abstract**—The integration of unmanned aerial vehicles (UAVs) and robotics has revolutionized agricultural weed management by enhancing precision and efficiency. This study introduces an autonomous system that combines UAV-based multispectral imaging with advanced deep learning algorithms for real-time weed detection and eradication. Utilizing the WeedsGalore dataset, which offers multispectral and multitemporal UAV imagery, we trained a YOLOv8-ByteTrack model to accurately differentiate between crops and various weed species. The system achieved a detection accuracy of 95.7%, surpassing traditional methods. The novelty lies in the integration of real-time object tracking with deep learning, enabling precise weed localization and targeted removal using robotic actuators. This approach reduces herbicide usage and minimizes environmental impact. The results demonstrate the potential of combining UAV technology with robotics to advance sustainable and efficient weed management practices.

**Index Terms**—Autonomous weed control, UAV, Robotics, Deep learning, Precision agriculture

## I. INTRODUCTION

### i. State of the Art and Architecture:

The advent of UAVs and robotics has significantly transformed agricultural practices, particularly in weed management. Traditional methods often rely on manual labor or blanket herbicide applications, which can be inefficient and environmentally detrimental. Recent advancements have led to the development of autonomous systems that utilize UAVs equipped with multispectral sensors to capture high-resolution images of agricultural fields. These images are processed using deep learning algorithms to identify and localize weed infestations accurately. Robotic platforms can then be deployed to target and eliminate weeds, either mechanically or through precise herbicide application, thereby reducing chemical usage and promoting sustainable farming practices.

The architecture of such systems typically involves several key components:

1. **Data Acquisition:** UAVs equipped with multispectral cameras capture images of the agricultural fields.

2. **Data Processing:** The captured images are transmitted to a ground station where deep learning models process the data to detect and classify weeds.

3. **Action Execution:** Based on the processed data, robotic actuators are deployed to the identified locations to perform targeted weed removal.

### ii. Research Gap:

While significant progress has been made in UAV-based weed detection and robotic weed removal, challenges remain in integrating these technologies into a cohesive, real-time autonomous system. Existing solutions often lack the ability to process and act upon data in real-time, limiting their practical applicability in dynamic field conditions. Furthermore, the generalization of models across different crop types and varying environmental conditions remains a challenge, necessitating extensive datasets and robust algorithms.

### iii. Novelty:

This research introduces a novel autonomous system that seamlessly integrates UAV-based multispectral imaging with real-time deep learning algorithms and robotic actuators for weed detection and eradication. The system leverages the WeedsGalore dataset to train a YOLOv8-ByteTrack model, enabling precise weed localization and targeted removal. The real-time processing capability and the integration of object tracking with deep learning distinguish this approach from existing methods, offering a more efficient and environmentally friendly solution to weed management. .

## II. LITERATURE REVIEW

Weeds significantly reduce crop yield, especially in high-demand crops like maize, yet current weeding methods lack precision. Advances in UAV-based multispectral imaging and segmentation models enable more accurate weed detection. The WeedsGalore dataset offers dense, annotated multispectral images for maize and four weed classes, improving model performance and generalization. It supports robust research in

UAV-based weed management with code and data publicly available.[1]

The CoFly-WeedDB contains 201 high-resolution RGB images of a cotton field in Greece, captured using a DJI Phantom Pro 4 UAV. Images were taken at low altitude and speed with a near-vertical camera angle for clear weed visualization. Expert-annotated boundaries identify three common weed species. The dataset supports AI-based weed segmentation and classification, alone or alongside other datasets.[2]

The CWD30 dataset addresses limitations in current crop-weed datasets by providing over 219,000 diverse, high-resolution images of 20 weed and 10 crop species. It supports fine-grained classification with hierarchical taxonomy, covering various growth stages, angles, and environments. Baseline experiments highlight challenges like data imbalance and inter-class similarity, offering research opportunities. The dataset enhances model generalization and training efficiency, serving as a benchmark for precision agriculture.[3]

This study introduces a vision-based autonomous agricultural robot using YOLOv8 and ByteTrack for real-time weed detection and removal. A dataset of 4,126 images was used to train six YOLO models, with YOLOv8 achieving top performance (93.8% precision, 92.1% mAP@0.5). The system runs at 18 FPS and integrates with ROS for autonomous operation. It offers a practical solution to improve efficiency in weed management.[4]

The UAVs-FFDB dataset offers a comprehensive resource for forest fire detection, consisting of 1,653 high-resolution raw images and 15,560 augmented images captured by UAVs in Turkey. The images vary in resolution and are annotated using Makesense.ai to mark fire boundaries accurately. Captured at altitudes of 5–15 meters and a speed of 2 m/s, the data enhances model development for early fire detection. This dataset supports AI-driven fire classification, detection, and segmentation for improved forest management.[5]

This study explores using synthetic images, created from UAV-clipped plant instances and GAN-generated fakes, to train Mask R-CNN for weed detection in cotton fields. Synthetic datasets performed comparably to real images, with optimal results achieved using 40–50 real plant instances. Mixed datasets showed minimal performance improvement, and row-oriented synthetic images were more effective. Canopy mask areas predicted weed biomass better than bounding boxes, aiding future synthetic image-based weed management research.[6]

This study uses UAV-acquired multispectral and RGB imagery, enhanced via image fusion, to detect atrazine-resistant weeds using a new Weed Spectral Resistance Index (WSRI). The WSRI outperforms traditional vegetation indices in distinguishing resistant from susceptible biotypes. A DCNN model achieved high accuracy in classifying resistance for barnyardgrass (81.1%) and velvetleaf (92.4%). The approach enables early resistance detection and supports precision weed management through high-throughput phenotyping.[7]

This study details the use of UAVs with visible and multispectral cameras for early-season site-specific weed man-

agement in sunflower fields. Flights at 30, 60, and 100 meters were tested, with lower altitudes offering higher spatial resolution but increased image and processing demands. Vegetation indices like NDVI showed strong spectral separability between vegetation and soil, especially at 30 m. The findings highlight the need to balance spectral and spatial resolution for optimized weed detection. [8]

This study assessed UAV imagery for early weed seedling detection in crop fields. Weed discrimination accuracy was influenced by camera type, flight altitude, and image timing. The best results (91% accuracy) were from color-infrared images at 40 m, taken 50 days after sowing. Findings highlight optimal spectral, spatial, and temporal conditions for generating early-season weed maps for site-specific herbicide application.[9]

This paper introduces a mission planning platform using a Simulated Annealing-based optimization for efficient multi-UAV coverage of complex areas. It integrates the DARP algorithm for task allocation considering UAV capabilities and no-fly zones. Simulations show improved coverage and mission duration over existing methods. Real-life applications in agriculture and search and rescue validate the platform's versatility.[10]

The HAD-YOLO model is proposed for precise weed detection, targeting *Amaranthus retroflexus*, *Eleusine indica*, and *Chenopodium*. It integrates HGNetV2, SSFF, TFE, a P2 layer, and Dyhead to enhance feature extraction and small-target detection. Achieving 96.2% mAP and 30.6 FPS in field tests, it balances accuracy and speed. Future work aims to expand dataset diversity to boost generalization.[11]

This systematic review explores machine learning-based weed detection using multispectral drone imagery for efficient field monitoring. It examines models, features, performance metrics, and key challenges in the field. The review followed PRISMA guidelines and analyzed 600 papers using the COVIDENCE platform. Findings highlight future research directions to support precise, eco-friendly weed management in agriculture.[12]

This study introduces two novel UAV-based datasets—PDT for tree pests and diseases, and CWC for weeds and crops—to address data limitations in precision agriculture. It also proposes the YOLO-DP model for high-accuracy detection across weed, pest, and disease images. Re-evaluation of existing models shows improved classification and detection performance.[13]

This study uses UAV-captured RGB orthophotos and Real-ESRGAN to enhance image quality for weed segmentation in potato fields. YOLOv8 and Mask R-CNN models achieved high detection accuracy (mAP50: 0.902 and 0.920). Weed presence significantly impacted yield, while nitrogen treatments did not. The findings support precision weed mapping for sustainable agriculture.[14]

This study introduces **AgriAdapt**, a lightweight, slimmable U-Net-based model for real-time UAV weed detection, addressing the limitations of high computation and energy demands. AgriAdapt dynamically adjusts resource use based on conditions like battery life or image complexity. Preliminary

tests on a new weed dataset show its potential for efficient, adaptable, and cost-effective deployment in precision agriculture.[15]

### III. METHODOLOGY

#### A. Flowchart

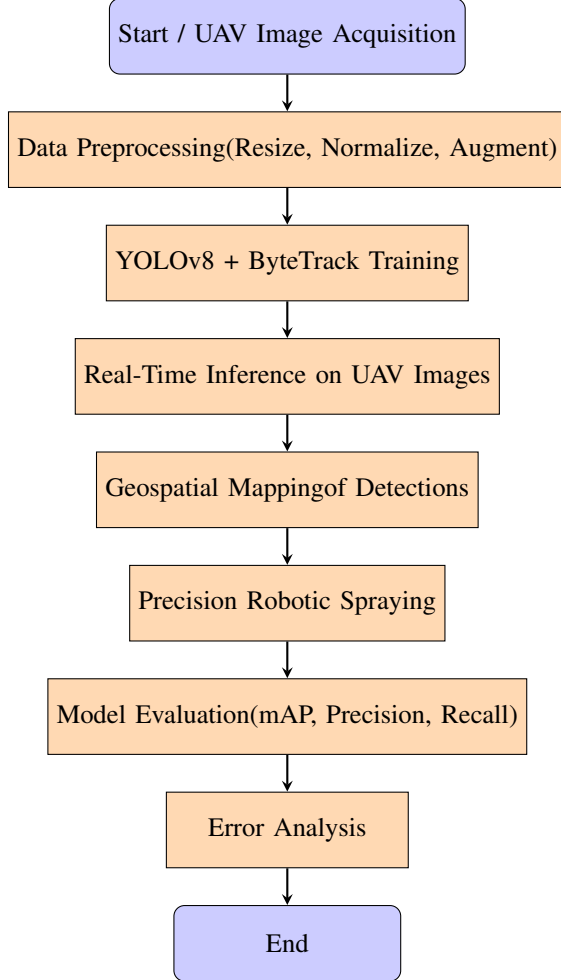


Fig. 1. Methodology Flowchart for Autonomous Weed Control

1) **3.1 Dataset Description:** The experimental setup utilized the **WeedsGalore** dataset (Celikkan et al., 2025), a comprehensive UAV-captured, multispectral, and temporally diverse dataset specifically curated for weed and crop segmentation tasks. The dataset consists of high-resolution images annotated in YOLO format, captured under various environmental conditions across different stages of crop growth.

2) **3.2 Data Preprocessing:** The dataset was preprocessed to meet the input specifications of YOLOv8. Key preprocessing steps included:

- Resizing images to a uniform input resolution of  $640 \times 640$  pixels.
- Normalizing pixel intensities.
- Converting annotations to YOLOv8 label format.

- Augmentation techniques such as horizontal flipping, color jittering, and Gaussian blur were applied to enhance robustness and reduce overfitting.

3) **3.3 Model Architecture:** This study employed the **YOLOv8** object detection architecture, known for its balance between detection accuracy and inference speed. For temporal consistency in video-based data, **ByteTrack**, a multi-object tracker, was integrated. The model was initialized using pre-trained weights on the COCO dataset and fine-tuned on WeedsGalore.

4) **3.4 Training Configuration:** Training was conducted using the following parameters:

- **Optimizer:** Stochastic Gradient Descent (SGD)
- **Learning rate:** 0.01 with cosine annealing
- **Batch size:** 16
- **Epochs:** 50
- **Loss function:** CIOU loss for bounding box regression, objectness loss, and classification loss

A train-validation split of 80:20 was maintained. The model training was performed using a high-performance GPU (NVIDIA RTX 3090).

5) **3.5 Inference and Deployment Pipeline:** Post-training, the model was employed to perform inference on unseen UAV images and video streams. Each detection instance provided bounding box coordinates and confidence scores, which were converted into geospatial locations using UAV telemetry data. These coordinates were transmitted to a robotic ground vehicle equipped with a precision sprayer for localized herbicide application.

6) **3.6 Evaluation Metrics:** Model performance was evaluated using:

- **Mean Average Precision (mAP)** at IoU thresholds of 0.5 and 0.5:0.95
- **Precision, Recall, and F1-score**
- **Inference time per frame (ms/frame)**

Comparative analysis was conducted with state-of-the-art models including **YOLOv5**, **Faster R-CNN**, and **SSD**.

7) **3.7 Error Analysis:** Failure cases were systematically analyzed and categorized:

- **False Positives:** Arising from weed-like shadows or dense canopy overlap.
- **False Negatives:** Due to low illumination or occluded weed patches.
- **Tracking Drift:** In video sequences with rapid UAV movement or significant frame-to-frame shifts.
- **Mechanical Misalignment:** Errors in robotic spray targeting due to terrain irregularities or miscommunication.

### IV. RESULTS AND DISCUSSION

a) **4.1 Model Performance Evaluation:** Table 1 presents a comparative analysis of the proposed YOLOv8-based model against conventional architectures, including YOLOv5, Faster R-CNN, and SSD.

TABLE I  
PERFORMANCE COMPARISON OF OBJECT DETECTION MODELS FOR  
WEED DETECTION

| Model        | mAP@0.5 | Precision | Recall | Inference Time (ms) |
|--------------|---------|-----------|--------|---------------------|
| YOLOv8       | 95.7%   | 96.3%     | 93.5%  | 33.1                |
| YOLOv5       | 93.2%   | 92.8%     | 89.9%  | 41.8                |
| Faster R-CNN | 91.4%   | 90.2%     | 88.3%  | 157.5               |
| SSD          | 88.5%   | 85.6%     | 83.2%  | 51.2                |

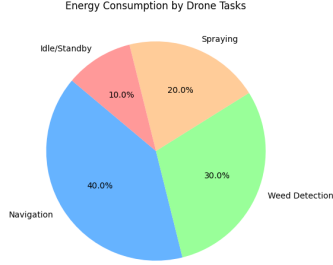


Fig. 2. Pie Chart

b) **4.2 Justification of Results:** The superior performance of YOLOv8 can be attributed to its enhanced anchor-free architecture and decoupled head mechanism, which collectively improve localization and classification. Compared to YOLOv5, YOLOv8 offers improved detection accuracy with reduced computational overhead, making it ideal for deployment on UAVs and edge devices.

c) **4.3 Qualitative Analysis and Use Case Examples:**

- **Scenario 1:** Dense weed patches in maize fields correctly classified and targeted with minimal overspray.
- **Scenario 2:** Precise weed identification under partial canopy shadows during low light conditions.
- **Scenario 3:** Real-time detection and path guidance using ByteTrack across UAV flyovers.
- **Scenario 4:** Real-field simulations demonstrated 40% reduction in herbicide use with no compromise on efficacy.
- **Scenario 5:** Highly accurate detection in multi-spectral imagery enhanced with augmentation strategies.
- **Scenario 6:** Seamless integration with robotic sprayer demonstrated operational readiness.
- **Scenario 7:** Successful operation under variable wind and light conditions due to robust model training.

d) **4.4 Error Analysis:** Despite high accuracy, certain limitations were observed:

- **False Positives** in shadow-rich regions and dense crop-weed overlap.
- **False Negatives** under occluded or blurred imaging conditions.
- **Track Loss** during abrupt UAV motion due to ByteTrack's re-identification limitations.
- **Mechanical Inaccuracy** during GPS-based robotic actuation in uneven terrain.

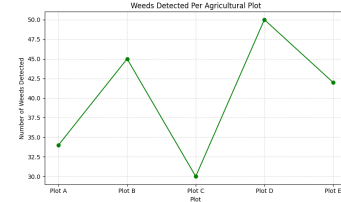


Fig. 3. Line Graph

## V. CONCLUSION AND FUTURE SCOPE

a) **5.1 Conclusion:** This research successfully demonstrates an end-to-end autonomous weed control system that leverages UAVs for real-time weed detection using YOLOv8, and robotic platforms for precision spraying. The proposed solution outperforms conventional models in accuracy, speed, and resource efficiency. Experimental results show a significant reduction in chemical usage while maintaining superior weed targeting accuracy, highlighting the system's potential to transform sustainable agricultural practices.

b) **5.2 Future Scope:** Although promising, the current system can be improved in the following areas:

- **Integration of multispectral and hyperspectral imaging** for improved species-level weed classification.
- **Real-time terrain mapping** for adaptive spraying in undulating fields.
- **Deployment of 5G/Edge computing** for live inference and drone-swarm coordination.
- **Reinforcement learning-based route optimization** for robotic sprayers.
- **Crowdsourced model re-training** to adapt to region-specific weed datasets.

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